Constraint Least Squares Estimate

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LSM

$$\underset{n\times 1}{Y} = \underset{n\times p}{X}\underset{p\times 1}{\beta} + \underset{n\times 1}{\varepsilon} \qquad \text{Constraint: } L^T\beta = \theta$$

Assume $\begin{cases} \text{rank}(X) = p \leq n \\ \text{rank}(L) = m \leq p \end{cases}$

Then the constrained LS estimator of β is

$$\tilde{\beta} = \underset{\beta:\, L^T\beta = \theta_0}{\text{argmin}} \|Y - X\beta\|_2^2 \text{ satisfies}$$

$$\tilde{\beta} = \hat{\beta} - (X^TX)^{-1}\left(L^T(X^TX)^{-1}L\right)^{-1}(L^T\hat{\beta} - \theta_0)$$

$\hat{\beta} \rightarrow$ unconstrained l.s. estimate

A transformation helps in simplifying computations

Define $\tilde{X} = X(X^TX)^{-1/2}$ $\quad$ $(X^TX)^{-1/2}$ is the symmetric square root of $(X^TX)^{-1}$

$\Rightarrow \tilde{X}^T\tilde{X} = I_p$

$\gamma = (X^TX)^{1/2}\beta$ so that $X\beta = \tilde{X}\gamma$ $\quad$ positive definite matrix

$\theta = L^T\beta = \tilde{L}^T\gamma$

where $\tilde{L} = L(X^TX)^{-1/2}$

Get $\tilde{\gamma} = \underset{\gamma:\, \tilde{L}^T\gamma = \theta_0}{\text{argmin}} \|Y - \tilde{X}\gamma\|^2$ —— (*)

$$\hat{\gamma} = \underset{\gamma}{\text{argmin}} \|Y - \tilde{X}\gamma\|^2 = \tilde{X}^TY$$

$\tilde{\beta} = (X^TX)^{-1/2}\tilde{\gamma} \qquad \hat{\beta} = (X^TX)^{-1/2}\hat{\gamma}$

Solve ⊛ by using Lagrange multiplier

$$f(\gamma, \lambda) = \|Y - \tilde{X}\gamma\|^2 + \lambda^T(\tilde{L}^T\gamma - \theta_0)$$

$$0 = \frac{\partial f}{\partial \gamma} = -2\tilde{X}^T(Y - \tilde{X}\gamma) + \tilde{L}\lambda \quad \text{———} \quad (i)$$

$$0 = \frac{\partial f}{\partial \gamma} = \tilde{L}^T\gamma - \theta_0 \quad \text{———} \quad (ii)$$

From (i)

$$2\underbrace{(\tilde{X}^T\tilde{X})}_{=I}\tilde{\gamma} = 2\tilde{X}^TY - \tilde{L}\lambda$$

$$\tilde{\gamma} = \tilde{X}^TY - \tfrac{1}{2}\tilde{L}\lambda$$

Plug into (ii) to solve for λ

$$0 = \tilde{L}^T\tilde{\gamma} - \theta_0 = \tilde{L}^T\left(\tilde{X}^TY - \tfrac{1}{2}\tilde{L}\lambda\right) - \theta_0$$

$$= \left(\tilde{L}^T\tilde{X}^TY - \theta_0\right) - \tfrac{1}{2}\underset{m\times p}{\tilde{L}^T}\underset{p\times m}{\tilde{L}}\lambda$$

$$\lambda = 2\left(\tilde{L}^T\tilde{L}\right)^{-1}\left(\tilde{L}^T\tilde{X}^TY - \theta_0\right)$$

$$= 2\left(\tilde{L}^T\tilde{L}\right)^{-1}\left(\tilde{L}^T\hat{\gamma} - \theta_0\right)$$

$$\tilde{\gamma} = \hat{\gamma} - \tilde{L}\left(\tilde{L}^T\tilde{L}\right)^{-1}\left(\tilde{L}^T\hat{\gamma} - \theta_0\right)$$

Multiply by $(X^TX)^{-1/2}$

$$\tilde{\beta} = \hat{\beta} - (X^TX)^{-1/2}(X^TX)^{-1/2}L\left(L^T(X^TX)^{-1/2}(X^TX)^{-1/2}L\right)^{-1}\left(L^T\hat{\beta} - \theta_0\right)$$

$$\tilde{\beta} = \hat{\beta} - (X^T X)^{-1} L \left(L^T (X^T X)^{-1} L\right)^{-1} \underbrace{(L^T \hat{\beta} - \theta_0)}_{(\hat{\theta} - \theta_0)}$$

Standard

$$\underset{\sim}{\varepsilon} \sim N(0, \sigma^2 I)$$

$$\hat{\beta} \sim N\left(\beta, \sigma^2 (X^T X)^{-1}\right)$$

$$\hat{\theta} = L^T \hat{\beta} \longleftarrow \text{BLUE of } \theta$$

$$\sim N\left(\underbrace{L^T \beta}_{\theta}, \sigma^2 L^T (X^T X)^{-1} L\right)$$

Corollary: $\underset{\|Y - X\tilde{\beta}\|^2}{\underset{\Downarrow}{SSE_{\text{constrained}}}} = \underset{\|Y - X\hat{\beta}\|^2}{\underset{\|}{SSE_{\text{full}}}} + (\hat{\theta} - \theta_0)^T \left(L^T (X^T X)^{-1} L\right)^{-1} (\hat{\theta} - \theta_0)$

Proof of corollary

$$\|Y - X\tilde{\beta}\|^2 = \|\underbrace{(Y - X\hat{\beta})}_{(I - P_X)Y} + \underbrace{(X\hat{\beta} - X\tilde{\beta})}_{\in \mathcal{C}(X)}\|^2$$

$$= \|Y - X\hat{\beta}\|^2 + \|X(\hat{\beta} - \tilde{\beta})\|^2$$

$$= \|Y - X\hat{\beta}\|^2 + (\hat{\theta} - \theta_0)^T \left(L^T (X^T X)^{-1} L\right)^{-1} L^T (X^T X)^{-1} \cancel{(X^T X)} (X^T X)^{-1} L \left(L^T (X^T X)^{-1} L\right)^{-1} (\hat{\theta} - \theta_0)$$

$$= SSE_{\text{full}} + (\hat{\theta} - \theta_0)^T \left(L^T (X^T X)^{-1} L\right)^{-1} (\hat{\theta} - \theta_0)$$

Application to inference under G-M framework

Linear hypothesis testing problem is to test

$$H_0 : L^T\beta = \theta_0 \ \underset{\text{(Reduced Model)}}{} \quad vs. \quad H_A : L^T\beta \neq \theta_0 \ \text{(Full model)}$$

Different Approaches:

1. Comparison of SSE (goodness of fit criteria)
2. Likelihood ratio test under the assumption by Gaussianity of errors.
3. Use studentized version of BLUE for $\theta = L^T\beta$

Based on the (distribution) of $\hat{\theta} = L^T\hat{\beta}$

$$W = (\hat{\theta}-\theta)^T s^2(\hat{\theta})^{-1}(\hat{\theta}-\theta) \qquad \text{Wald's Test Statistic}$$

Claim: All three approaches are equivalent.

① ≡ ③ : Test Statistic under approach I is

$$\frac{SSE_{\underset{(constrained)}{reduced}} - SSE_{full}}{SSE_{full}}$$

by the corollary

$$= \frac{(\hat{\theta}-\theta_0)^T\left(L^T(X^TX)^{-1}L\right)^{-1}(\hat{\theta}-\theta_0)}{(n-p)\hat{\sigma}^2}$$

$$MSE_{full} = \frac{SSE_{full}}{n-p}$$

$$= \frac{W}{n-p} \qquad s^2(\hat{\theta}) = \hat{\sigma}^2\left(L^T(X^TX)^{-1}L\right)$$

Likelihood Test

Compute MLE without constraint i.e. when $\beta \in \mathbb{R}^p$ and MLE under constraint $H_0: L^T\beta = \theta_0$

log likelihood $(\underset{\sim}{\beta}, \sigma^2)$

$$c_n \left[-\frac{1}{2\sigma^2} \|Y - X\beta\|^2 - \frac{n}{2}\log\sigma^2 \right] = g(\underset{\sim}{\beta}, \sigma^2)$$

Unconstrained MLE $\longleftrightarrow \min_{\beta} \|Y - X\beta\|^2$

$$\hat{\beta}_{MLE} = \beta, \quad \hat{\sigma}^2_{MLE} = \frac{SSE_{full}}{n} = \frac{\|Y - X\hat{\beta}\|^2}{n}$$

Constrained MLE

$$(\tilde{\beta}, \tilde{\sigma}^2) = \underset{\beta : L^T\beta = \theta_0}{\arg\max}\, g(\beta, \sigma^2)$$

$$\Leftrightarrow \underset{\beta : L^T\beta = \theta_0}{\arg\min} \|Y - X\beta\|^2 \quad (\text{fixed } \sigma^2)$$

$\tilde{\beta}$ = constrained LSE of β

$$\tilde{\sigma}^2 = \frac{\|Y - X\tilde{\beta}\|^2}{n} = \frac{SSE_{red}}{n}$$

LR statistic for testing $H_0: L^T\beta = \theta_0$ vs. $H_A: L^T\beta \neq \theta_0$ is

$$\frac{\sup_{\underset{\sim}{\beta}: L^T\beta = \theta_0,\ \sigma^2 > 0} f_{\beta,\sigma^2}(Y|X)}{\sup_{\underset{\sim}{\beta} \in \mathbb{R}^p,\ \sigma^2 > 0} f_{\beta,\sigma^2}(Y|X)}$$

$$= \frac{\frac{1}{(\tilde{\sigma}^2)^{n/2}} e^{-\frac{1}{2\tilde{\sigma}^2}\|Y - X\tilde{\beta}\|^2}}{\frac{1}{(\hat{\sigma}^2)^{n/2}} e^{-\frac{1}{2\hat{\sigma}^2}\|Y - X\hat{\beta}\|^2}}$$

$$= \left(\frac{\hat{\sigma}^2}{\tilde{\sigma}^2}\right)^{n/2} = \left(\frac{SSE_{full}}{SSE_{red}}\right)^{n/2}$$

$$= \frac{1}{\left(1 + \left(\frac{SSE_{red} - SSE_{full}}{SSE_{full}}\right)\right)^{n/2}}$$

Recall the problem

$$Y = \mathbb{X}\beta + \varepsilon \qquad \underset{n \times p}{\mathbb{X}} = [\underset{n \times q}{\mathbb{Z}} \ \underset{n \times r}{\mathbb{W}}]$$

$$\beta = \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \qquad p = \text{rank}(\mathbb{X})$$

$$= \text{rank}(\mathbb{Z}) + \text{rank}(\mathbb{W})$$

$$= q + r$$

Test: the submodel

$Y = \mathbb{Z}\gamma + \varepsilon$ is adequate

$\Leftrightarrow H_0 : \delta = 0$ vs. $H_A : \delta \neq 0$

$$\delta = L^T\beta = (0 \ I)\begin{pmatrix} \gamma \\ \delta \end{pmatrix} \qquad L_{p \times r} = \begin{pmatrix} 0_{q \times r} \\ I_{r \times r} \end{pmatrix}$$

Wald's approach: $\underset{m \times 1}{\hat{\theta}} = L^T\hat{\beta} \sim N(L^T\beta, \sigma^2(L^T(X^TX)^{-1}L))$

under H_0: $$\frac{(\hat{\theta} - \theta_0)(L^T(X^TX)^{-1}L)^{-1}(\hat{\theta} - \theta_0)}{\sigma^2} \sim \chi^2_m$$

$$\left[\begin{array}{l} \hat{\theta}_{m\times 1} \sim N(\theta_0, \Sigma) \\ \Sigma^{-1/2}(\hat{\theta}-\theta_0) \sim N(0, I) \\ (\hat{\theta}-\theta_0)^T \Sigma^{-1}(\hat{\theta}-\theta_0) \sim \chi^2_m \end{array}\right]$$

We know $\hat{\theta}$ is a Linear Transformation of $\hat{\beta}$

and $\hat{\sigma}^2 = \|(I-P_X)Y\|^2/(n-p)$ — independently distributed

$$(I-P_X)Y = \underbrace{(I-P_X)X\beta}_{0} + (I-P_X)\varepsilon$$

$$= (I-P_X)\varepsilon$$

$$\|(I-P_X)Y\|^2 = \|(I-P_X)\varepsilon\|^2$$

$$= \sigma^2\Big[\underbrace{\left(\frac{\varepsilon}{\sigma}\right)^T}_{U} \underbrace{(I-P_X)}_{\text{rank}=(n-p)} \left(\frac{\varepsilon}{\sigma}\right)\Big] \qquad U \sim N(0, I_n)$$

$$\sim \sigma^2 \chi_{n-p}$$

Wald's statistic

$$\frac{(\hat{\theta}-\theta_0)^T (L^T(X^TX)^{-1}L)^{-1}(\hat{\theta}-\theta_0) \times \frac{1}{\sigma^2}}{\hat{\sigma}^2 \times \frac{1}{\sigma^2}}$$

$$\stackrel{\mathcal{D}}{=} \frac{\chi^2_m}{\chi^2_{n-p}/(n-p)} = mF_{m,n-p}$$